

Let the Sheet Do the Work — Paper A

MEDIUM • NO COMPUTER • NO CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. There is no computer and no calculator: every number is designed to work by hand.
- This paper is about **Google Sheets**. Every formula starts with **=**, arguments are separated by **commas**, and text goes in **quotation marks**.
- Where a question asks for a **value**, write what the cell would show. Where it asks for a **formula**, write it exactly, brackets and commas included.
- The mark for each question is shown in square brackets on the right.
- **Reminder (medium papers only).** SUM(range) AVERAGE(range) MIN(range) MAX(range) COUNT(range) COUNTA(range) MINUS(a, b) DIVIDE(a, b) PRODUCT(a, b) IF(test, if_true, if_false) SUMIF(range, criterion, sum_range) COUNTIF(range, criterion)

SECTION 1 – READ THE SHEET

1. The tuck-shop sheet below is open in Google Sheets. Column D holds a formula in every row; the extract shows each one as typed. Write the value each of these would show.

	A	B	C	D
1	Item	Price (₹)	Sold	Revenue (₹)
2	Samosa	15	12	=B2*C2
3	Vada Pav	20	11	=B3*C3
4	Poha	30	8	=B4*C4
5	Chai	10	15	=B5*C5
6	Lassi	25	9	=B6*C6

(a) The cell D5 [1]

(b) =SUM(D2:D6) [1]

(c) =MAX(D2:D6) [1]

(d) =AVERAGE(C2:C6) [1]

2. Eight students sat a test out of 50. Two were absent, so the teacher typed the word **absent**, and one cell was left empty. Write the value of each.

	A	B
1	Student	Test mark (out of 50)
2	Aditi	34
3	Bhavna	41
4	Chetan	absent
5	Deepa	28
6	Eshan	
7	Farah	45
8	Girish	absent
9	Hema	32

(a) =COUNT (B2:B9) [1]

(b) =COUNTA (B2:B9) [1]

(c) =AVERAGE (B2:B9) [1]

(d) Mr Rao wants to know how many students actually sat the test. Say which of COUNT and COUNTA he should use, and why. [1]

3. Column C holds this formula in C2, filled down: =IF (B2>=40, "Pass", "Fail") [4]
Write what each of C2, C3, C4 and C5 shows.

	A	B	C
1	Student	Score	Result
2	Anil	47	
3	Bina	40	
4	Charu	39	
5	Dev	12	

4. A class sheet has Class in column B and Marks in column C, rows 2 to 8.

	A	B	C
1	Name	Class	Marks
2	Farhan	A	44
3	Gauri	B	31
4	Harini	A	29
5	Imran	B	37
6	Jatin	A	38
7	Kavya	B	46
8	Lakshman	A	22

- (a) Write a formula that shows the highest mark. [1]

- (b) Write a formula that counts the students in Class A. [1]

- (c) Write a formula that adds up the marks of the Class B students only, and state the value it shows. [2]

Write the formula, then the value. A value with no formula next to it will not earn the formula mark.

5. A mark out of 50 is typed in cell C2. Column D should show **High** for 30 or more, **Medium** for 20 to 29, and **Low** for below 20.

- (a) Write the formula for D2, using a nested IF. [3]

- (b) State what your formula shows when C2 is exactly 20. [1]

6. A school trip sheet holds the cost of the bus in A2, the number of students in B2, and the money collected so far in C2.

	A	B	C
1	Bus (₹)	Students	Collected (₹)
2	2400	30	1800

- (a) Using DIVIDE, write the formula for the cost per student, and state its value. [1]

- (b) Using MINUS, write the formula for the money still needed, and state its value. [1]

SECTION 3 – SORT, FILTER AND PROTECT

7. A student wants to put this table in order of score, smallest first. She selects **only** the Score cells, B2 to B6, and uses Data > Sort range, A to Z. She does not select column A.

	A	B
1	Name	Score
2	Dev	14
3	Esha	22
4	Fatima	9
5	Gopal	31
6	Hari	18

- (a) Write what the table holds after her sort. [2]

	A	B
1	Name	Score
2		
3		
4		
5		
6		

- (b) Dev really scored 14. What does the sheet now say Dev scored? [1]

- (c) Write the names in the order they should appear after a **correct** sort, smallest score first. [1]
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8. A filter is applied to this table with the condition **Marks is greater than or equal to 35**.

	A	B
1	Subject	Marks
2	Maths	42
3	Science	28
4	English	36
5	Maths	31
6	Science	35
7	English	19

- (a) Write the row numbers that are still showing. [1]
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- (b) Row 3 is no longer on the screen. Say exactly what has happened to it. [1]
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9. A column holds marks out of 50. Design a data validation rule that stops anyone entering a mark above 50 or below 0.

- (a) State the criteria, with the numbers. [1]
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- (b) State the setting for invalid data, and why you chose it. [1]
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End of paper. Check that every formula starts with = and that every word inside a formula is in quotation marks.